Madhav Malpani

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EDUCATION

University of California San Diego

- Bachelor of Science in Artificial Intelligence *Expected graduation date: June 2030*

Delhi Public School (*Indian High School, Central Board of Secondary Education***)**

- Class 12 Final Exam (Mathematics, Physics, Chemistry, Computer Science, English): 86.7% *May 2026*
- Class 10 Final Exam (Mathematics, English, Science, Social Studies, Hindi): 95.6%. *May 2024*

EXPERIENCE

***Mechanical Design and Fabrication Intern* |** *Jun 2025 – Jul 2025 | Surat, Gujarat*
Space Technology and Aeronautical Rocketry (Space-Tech/Research Startup)
(LinkedIn)

- Simulated EMI radiation from a steel tape antenna for a 1U CubeSat using CST Studio
- Designed and soldered the PCB for the GPS and communication system of the CubeSat.
- Sketched possible burn-wire antenna deployment mechanisms inspired from Japanese wood joinery.
- Presented the CubeSat's capabilities and applications, engaging founders and 15+ trainees

***Machine Learning Trainee* |** *Jan 2025 – Feb 2025 | Hybrid*
Indian Institute of Technology Kanpur, Techkriti (Student led organization)

- Studied fundamental machine learning models & linear algebra concepts to understand the foundations of ML.
- Applied these concepts by developing a digit classification project using SVM model that reached 98 percent accuracy

Electronics Sub-Team recruit | *Aug 2026 – Present*
Triton Robotics, University of California San Diego

- Completing the 4-week mandatory recruit training. Modified the old centerboard for a robot by designing a new circuit board.

PROJECTS

Quadruped | *Aug 2026 – Present* (GitHub)

- Assembled an open-sourced quadruped to understand the code & mathematics behind the forward and inverse kinematics.
- Coded a python framework for controlling servos according to input joint angles and end-effector coordinates.

Air Quality Monitor | *Sep 2025 – Nov 2025* (GitHub) (LinkedIn)

- Built a real-time air quality monitor to track the live concentration of pollutants in my surroundings.
- Developed a dual-breadboard prototype to learn circuit construction and component interactions.
- Analyzed manufacturer datasheets and integrated power supply system according to the operating values of components.
- Used AI developer tools within the Arduino IDE to test, and debug firmware for the graphical user interface on the display.

HydroSafe | *Aug 2025 – Oct 2025* (LinkedIn)

- Built a working prototype of a tile integrating piezoelectric sensors, springs for storing for enduring a vehicle's compression and pipes for water drainage to find a solution to water clogged roads and potholes on roads.
- Collaborated with a classmate to form hypothetical failure situations and find a range for spring's constant.
- Presented the project at a local science fair to spread awareness amongst more than 100 visitors including the mayor.

AWARDS & HONOURS

International Computer Science Competition Finalist 2025

- Involved MCQs based and Research Paper Analysis based tests on computer science topic

Mental Math World Cup 2025: Ranked 411 out of 8,600 globally

- International competition focused on speed and accuracy in mental arithmetic

Ramanujan National Math Competition 2024: Nationally ranked 373

- National-level mathematics competition testing problem-solving and mathematical reasoning

SKILLS

Software Tools: Onshape, GitHub, EasyEDA, CST Studio, Visual Studio Code, Microsoft Excel, Microsoft Word, Google Sheets
Programming Languages: Python
Engineering & Technical: Embedded Systems, PCB Design, Soldering, IoT Systems, AI-assisted Programming
Training Programs: IIT Kanpur Python for AIML Certification, IIT Kanpur Techkriti ML Winter Program, IIT Bombay ML Workshop

PROJECT PORTFOLIO

PERSONAL PROJECTS